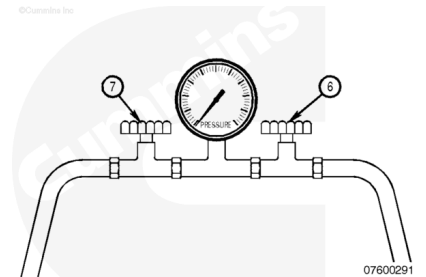


Trainings ▾

Test

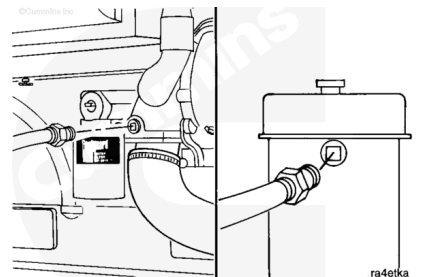
To check for a radiator obstruction perform the following steps.

Use a pressure gauge designed to measure 0 to 100 kPa [0 to 15 psi]. Install a valve on each side of the gauge.

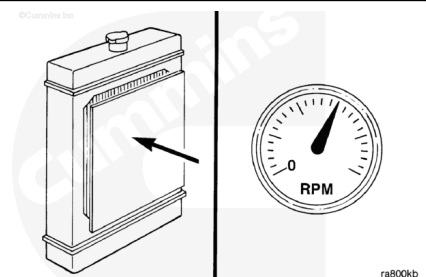


Install one line into the water pump inlet connection and the other into the thermostat return line on top of the radiator core.

Bleed the air from the gauge lines.



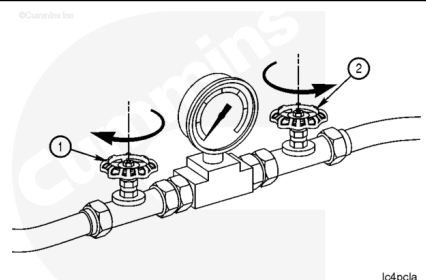
Restrict the radiator airflow. Operate the engine until the coolant temperature reaches 88 to 90°C [190 to 195°F].



Operate the engine at rated rpm.

Close valve (1), and open valve (2).

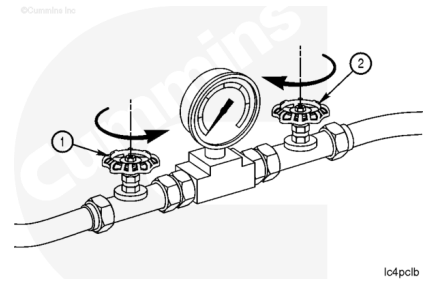
Record the pressure at the top of the radiator.



Close valve (2) and open valve (1).

Record the pressure at the water pump inlet.

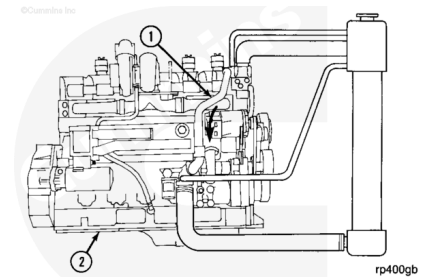
Compare the two pressure readings. If they differ by more than 35 kPa [5 psi], the radiator core or piping is obstructed.



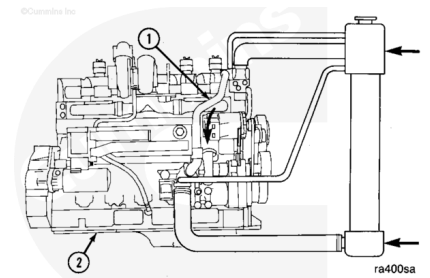
To perform a reverse flow test perform the following steps.

This test **must** be performed with the cooling system completely full and the coolant at ambient temperature.

Operate the engine at medium to high idle. Since the thermostats are closed, coolant flow is through the bypass line (1) and the engine (2).

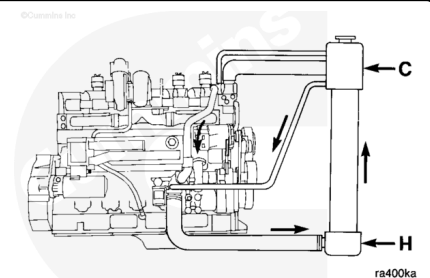


As the engine is warming up, use a contact pyrometer to check the temperature of the radiator top and bottom tanks.



If the bottom tank begins to warm and the top tank remains cold, the radiator has reverse flow. Refer to the manufacturer's instructions for repair procedures.

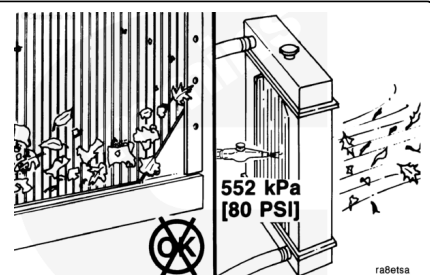
Check the piping to be sure it is the correct size and routed according to specifications. Refer to the appropriate installation recommendation bulletin.



Finishing Steps

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Check for plugged radiator fins.

Use compressed air to blow out the dirt and debris.

Blow the compressed air opposite the direction of the fan airflow.

Measurements

	kpa	psi
Air Pressure:	552	80

Check for bent or broken fins. Check for radiator core and gasket leaks.

If the radiator **must** be replaced, refer to the OEM's replacement procedures.

